Solving Multi-Step Equations: 50 Problems (Easy → Hard)

Solve for x in each equation. Problems are arranged by difficulty: two-step equations, then variables on both sides, then equations with distribution, then equations with fractions/decimals, then mixed equations with grouping. An answer key is provided at the end.

Level 1 — Two-step equations (#1–10)

1. $2x + 3 = 11$
2. $3x - 5 = 10$
3. $5x + 2 = 17$
4. $4x - 7 = 13$
5. $-2x + 6 = 12$
6. $\frac{x}{3} + 4 = 7$
7. $\frac{x}{2} - 1 = 5$
8. $6 - 3x = 0$
9. $-3x + 8 = -7$
10. $\frac{2x}{5} = 6$

Level 2 — Variables on both sides & simple distribution (#11–20)

11. $2x + 3 = x + 7$
12. $5x - 4 = 3x + 6$
13. $7x + 1 = 4x + 16$
14. $3(x + 2) = 15$
15. $2(x - 3) = 10$
16. $4(2x - 1) = 20$
17. $-3(x + 5) = 6$
18. $5 = 2(x + 1)$
19. $3x + 7 = 2x + 10$
20. $8 - x = 2x + 2$

Level 3 — Distribute + combine + variables on both sides (#21–30)

21. $3(x + 4) = 2x + 18$
22. $2(x - 1) + 3x = 18$
23. $5x - 2(x + 3) = 9$
24. $4(x + 2) - 3 = 17$
25. $2(3x - 1) + 4 = 20$
26. $4x + 7 - 2x = 19$
27. $3(x - 2) = 2(x + 1)$
28. $5(x + 1) = 3(x - 1) + 12$

29. $-2(x-4) = 3x - 7$

30. $7 - 2(x+1) = 3x$

Level 4 — Fractions and decimals (#31–40)

31. $\dfrac{x}{2} + \dfrac{x}{3} = 10$

32. $\dfrac{2x}{3} - 4 = 6$

33. $\dfrac{x-1}{2} = 4$

34. $\dfrac{x+3}{4} = 2x - 5$

35. $\dfrac{3x-2}{5} = \dfrac{x+4}{2}$

36. $0.4x + 1.2 = 3.6$

37. $0.5(x+4) = 0.25x + 5$

38. $\dfrac{x}{2} + \dfrac{x}{4} - 3 = 5$

39. $\dfrac{2x-1}{3} + 5 = 8$

40. $\dfrac{x+5}{3} - \dfrac{x-1}{2} = 1$

Level 5 — Multi-step with grouping (#41–50)

41. $3(2x-4) - 5(x-1) = 7$

42. $2(x+3) - 4(x-1) = -6$

43. $-(x-3) + 2(2x+1) = 4x + 8$

44. $5x - 3(2-x) = 4(x+1) - 6$

45. $4(x-1) - 2(3x+2) = -10$

46. $2[\,3 - (x+1)\,] = x - 4$

47. $3[\,2x - (x+1)\,] = 5(x-3) - 2$

48. $\dfrac{1}{2}(x-4) + \dfrac{1}{3}(x+6) = 7$

49. $\dfrac{3}{4}(2x-1) - \dfrac{1}{2}(x+5) = \dfrac{1}{4}(x+3)$

50. $2(x+3) - \big[\,3 - 2(x-1)\,\big] = 3(x+1)$

Answer Key

1. $x = 4$
2. $x = 5$
3. $x = 3$
4. $x = 5$
5. $x = -3$
6. $x = 9$
7. $x = 12$
8. $x = 2$
9. $x = 5$
10. $x = 15$
11. $x = 4$
12. $x = 5$
13. $x = 5$
14. $x = 3$
15. $x = 8$
16. $x = 3$
17. $x = -7$
18. $x = \frac{3}{2}$
19. $x = 3$
20. $x = 2$
21. $x = 6$
22. $x = 4$
23. $x = 5$
24. $x = 3$
25. $x = 3$
26. $x = 6$
27. $x = 8$
28. $x = 2$
29. $x = 3$
30. $x = 1$
31. $x = 12$
32. $x = 15$
33. $x = 9$
34. $x = \frac{23}{7}$
35. $x = 24$
36. $x = 6$
37. $x = 12$
38. $x = \frac{32}{3}$
39. $x = 5$
40. $x = 7$
41. $x = 14$
42. $x = 8$
43. $x = -3$
44. $x = 1$
45. $x = 1$
46. $x = \frac{8}{3}$
47. $x = 7$
48. $x = \frac{42}{5}$
49. $x = \frac{16}{3}$
50. $x = 2$

Tips for multi-step equations.

1. Clear parentheses by distributing.
2. Clear fractions by multiplying both sides by the LCD; clear decimals by multiplying by a power of 10.
3. Combine like terms on each side.
4. Move variable terms to one side, constants to the other.
5. Divide (or multiply) to isolate x.
6. Substitute your answer back into the original equation to check.